# Problem 5

**Problem.** For the boundary value problem with periodic boundary conditions

$$\begin{aligned} \textbf{ODE}: &\quad -u'' + q(x)\,u = f(x), \\ \textbf{BC1}: &\quad u(-\pi) = u(\pi), \\ \textbf{BC2}: &\quad u'(-\pi) = u'(\pi), \end{aligned}$$

where $q > 0\ \forall x \in [-\pi, \pi]$:

Numerically implement the CG(1) method on a uniform mesh with the following functions

$$q(x) = 3 - \sin(x) - \sin(2x) - \cos(x), \quad f(x) = 2e^{\sin(x)}e^{\cos(x)}, \quad u(x) = e^{\sin(x)}e^{\cos(x)}.$$

Numerically determine the convergence rate in both the *energy norm* and the $L^2$-norm by creating an appropriate convergence table with

$$M = 24, 49, 99, 199, 399, 799.$$

**HINT:** You can assume a uniform grid spacing $h = 2\pi/\,(M+1)$. For a given $M$, you will need to write Python code that assembles the stiffness matrix, $\underline{\underline{A}}$, and the right-hand side vector, $\underline{b}$. I recommend that you store $\underline{\underline{A}}$ as a sparse array using `scipy.sparse` and that you solve the resulting linear system using the built-in conjugate gradient solver `cg` from `scipy.sparse.linalg`. You can find documentation for how to use these functions here: https://docs.scipy.org/doc/scipy/.

## Verification of the data

With $u = e^{\sin x + \cos x}$: $u' = u(\cos x - \sin x)$ and $u'' = u\big[(\cos x - \sin x)^2 - (\sin x + \cos x)\big]$. Since $(\cos x - \sin x)^2 = 1 - \sin 2x$,

$$-u'' + q\,u = u\big[-1 + \sin 2x + \sin x + \cos x + 3 - \sin x - \sin 2x - \cos x\big] = 2u = f. \tag{30}$$

Numerically $\max|-u'' + qu - f| = 2.7 \times 10^{-15}$, and $|u(-\pi) - u(\pi)| + |u'(-\pi) - u'(\pi)| = 3.3 \times 10^{-16}$, so $u$ satisfies both boundary conditions.

## Implementation

- Mesh and basis as in Problem 2: $N = M + 1$ unknowns, $N$ elements, $h = 2\pi/N$.
- Element contributions to (16), on the reference element $[0, 1]$ with $N_0(\xi) = 1 - \xi$, $N_1(\xi) = \xi$:

$$\boldsymbol{K}^e_{\text{stiff}} = \frac{1}{h}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}, \qquad K^e_{ab,\,\text{mass}} = \int_{I_e} q\,N_a N_b\,\mathrm{d}x, \qquad b^e_a = \int_{I_e} f\,N_a\,\mathrm{d}x. \tag{31}$$

  The stiffness block is exact; the $q$-weighted mass block and the load are evaluated by 6-point Gauss–Legendre quadrature, which is far more accurate than the $O(h^2)$ discretisation error.

- Local-to-global map, the one place periodicity enters:

$$\text{element } e \;\longmapsto\; \text{nodes } \big(e,\ (e+1) \bmod N\big), \qquad e = 0, \ldots, N-1. \tag{32}$$

  The $\bmod N$ returns the last element's right node to node 0, producing the corner entries $K_{0,M} = K_{M,0}$ of Problem 2. No row or column is deleted.

- Solved with `scipy.sparse.linalg.cg` at tolerance $10^{-13}$, admissible because $\boldsymbol{K}$ is symmetric positive definite (Problem 2).
- Both error norms are computed element by element with the same Gauss rule, using $u'$ analytically and $U'$ constant on each element.

## Convergence table

Observed rate between consecutive meshes:

$$r_k = \frac{\ln\big(E_{k-1}/E_k\big)}{\ln\big(h_{k-1}/h_k\big)}. \tag{33}$$

| $M$ | $h$ | $\|u-U\|_{L^2}$ | rate | $\|u-U\|_E$ | rate |
|---|---|---|---|---|---|
| 24 | 2.5133e-01 | 2.7134e-02 | — | 4.1346e-01 | — |
| 49 | 1.2566e-01 | 6.7860e-03 | 2.00 | 2.0691e-01 | 1.00 |
| 99 | 6.2832e-02 | 1.6967e-03 | 2.00 | 1.0348e-01 | 1.00 |
| 199 | 3.1416e-02 | 4.2418e-04 | 2.00 | 5.1743e-02 | 1.00 |
| 399 | 1.5708e-02 | 1.0605e-04 | 2.00 | 2.5872e-02 | 1.00 |
| 799 | 7.8540e-03 | 2.6511e-05 | 2.00 | 1.2936e-02 | 1.00 |

## Conclusion

$$\|u-U\|_E = O(h^{1.00}), \qquad \|u-U\|_{L^2} = O(h^{2.00}),$$

confirming (22) (first order in the energy norm) and (29) (second order in $L^2$).

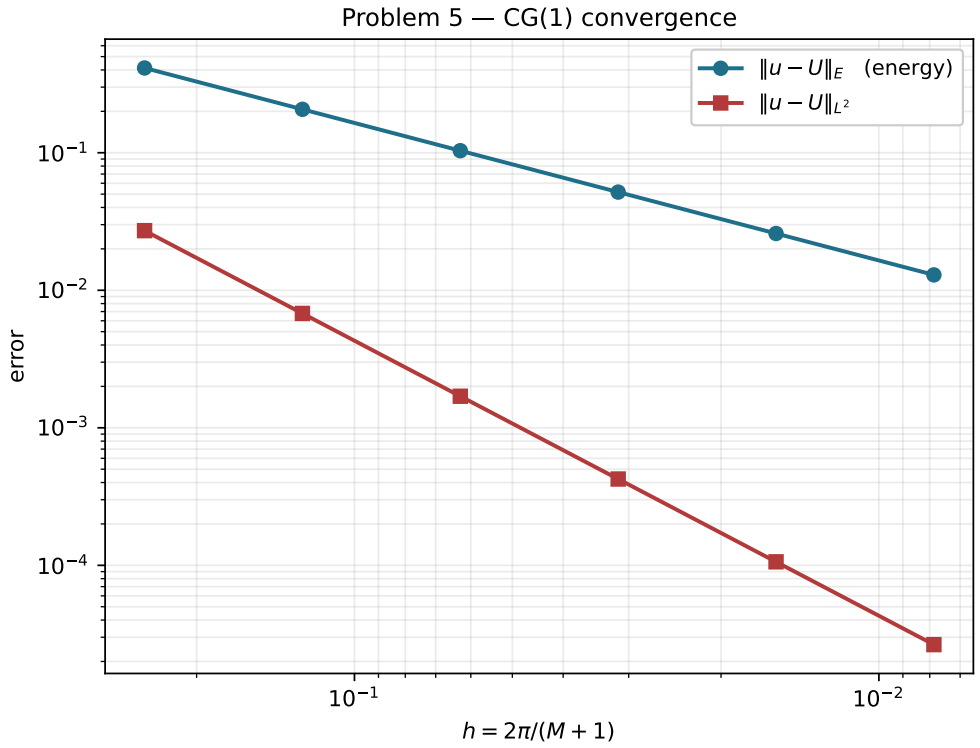



Figure 1: Problem 5. CG(1) convergence for the periodic problem: energy-norm and $L^2$-norm errors against the mesh size $h = 2\pi/(M+1)$, for $M = 24, 49, 99, 199, 399, 799$. Both curves are straight on log–log axes, with measured slopes 1.00 and 2.00.

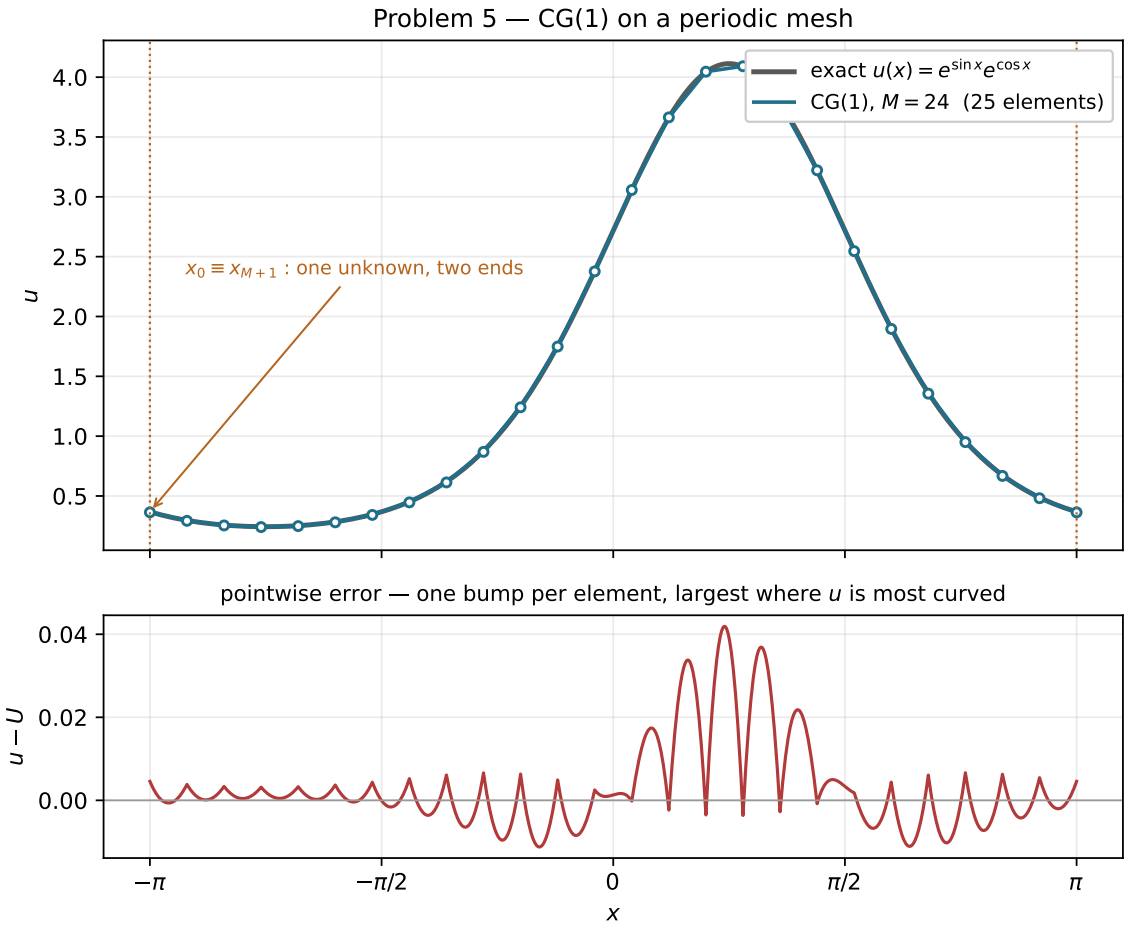



Figure 2: Problem 5. **Top:** exact solution and the CG(1) solution at $M = 24$; the circles are the 25 nodes, and the dotted lines at $x = \pm\pi$ mark the same point of the domain, one unknown $\xi_0$ serving both ends. **Bottom:** the pointwise error $u - U$, one bump per element and largest where $u$ is most strongly curved.

# Problem 6

**Problem.** Implement the CG(2) method (piecewise quadratics) as discussed in class for $-u'' = f$ on $(0,1)$, $u(0) = u(1) = 0$.

## Mesh and numbering

- $h = 1/(M+1)$, $M+1$ elements, nodes indexed by half-integers:

$$x_{k/2} = \tfrac{kh}{2}, \quad k = 0, 1, \ldots, 2M+2: \qquad k \text{ even} \Rightarrow \text{vertex}, \qquad k \text{ odd} \Rightarrow \text{midside}. \tag{34}$$

- $2M+3$ nodes; the two Dirichlet ends $k = 0$ and $k = 2M+2$ are eliminated, leaving $2M+1$ unknowns — the size Lecture 10 gives.

## Basis and element stiffness

- On the reference element $[0,1]$:

$$N_0(\xi) = 2\big(\xi - \tfrac{1}{2}\big)(\xi - 1), \qquad N_1(\xi) = 4\xi(1-\xi), \qquad N_2(\xi) = 2\xi\big(\xi - \tfrac{1}{2}\big), \tag{35}$$

  equal to 1 at $\xi = 0, \frac{1}{2}, 1$ respectively and 0 at the other two.

- With $\mathrm{d}N_a/\mathrm{d}x = h^{-1}\mathrm{d}N_a/\mathrm{d}\xi$ and $\mathrm{d}N_0/\mathrm{d}\xi = 4\xi - 3$, $\mathrm{d}N_1/\mathrm{d}\xi = 4 - 8\xi$, $\mathrm{d}N_2/\mathrm{d}\xi = 4\xi - 1$, the element stiffness integrates exactly to

$$\boldsymbol{K}^e = \frac{1}{3h}\begin{bmatrix} 7 & -8 & 1 \\ -8 & 16 & -8 \\ 1 & -8 & 7 \end{bmatrix} \qquad \text{(ordering: vertex, midside, vertex).} \tag{36}$$

## Load vector

- Assembled from the exact $P_2$ element mass matrix applied to nodal $f$:

$$\boldsymbol{b}^e = h\,\boldsymbol{M}^e\boldsymbol{f}^e, \qquad \boldsymbol{M}^e = \frac{1}{30}\begin{bmatrix} 4 & 2 & -1 \\ 2 & 16 & 2 \\ -1 & 2 & 4 \end{bmatrix}, \qquad \boldsymbol{f}^e = \big(f(x_{j-1}),\, f(x_{j-1/2}),\, f(x_j)\big)^T, \tag{37}$$

  which reproduces the two lecture stencils row by row.

- Exact quadrature of $\int f N_a \,\mathrm{d}x$ was also implemented as a check; it changes the asymptotic rates not at all, and the table entries only in the third digit at $M = 24$ and 49.

## Verification against Lecture 10

Assembling (36) and striking out the two Dirichlet rows and columns, the code prints $3h\boldsymbol{K}$ for $M = 4$ (a $9 \times 9$ system):

$$3h\,\boldsymbol{K} = \begin{bmatrix} 16 & -8 & & & & & & & \\ -8 & 14 & -8 & 1 & & & & & \\ & -8 & 16 & -8 & & & & & \\ & 1 & -8 & 14 & -8 & 1 & & & \\ & & & -8 & 16 & -8 & & & \\ & & & 1 & -8 & 14 & -8 & 1 & \\ & & & & & -8 & 16 & -8 & \\ & & & & & 1 & -8 & 14 & -8 \\ & & & & & & & -8 & 16 \end{bmatrix} \tag{38}$$

This reproduces Lecture 10 exactly: $16/3h$ on the midside diagonal, $14/3h$ on the vertex diagonal, $-8/3h$ on the first off-diagonal, $1/3h$ two places out from a vertex row. The matrix is symmetric positive definite, so `cg` applies; its pattern is pentadiagonal rather than tridiagonal.

## Problem 7

**Problem.** Apply the method to

$$u(x) = e^{1-\cos(2\pi x)-\sin(2\pi x)} - 1, \qquad f(x) = 4\pi^2 e^{1-\cos(2\pi x)-\sin(2\pi x)}\big[\sin(4\pi x) - \cos(2\pi x) - \sin(2\pi x) - 1\big],$$

and determine the convergence rate in both norms for $M = 24, 49, 99, 199, 399, 799$.

### Verification of the data

Write $g = 1 - \cos 2\pi x - \sin 2\pi x$, so $u = e^g - 1$ and

$$g' = 2\pi(\sin 2\pi x - \cos 2\pi x), \qquad g'' = 4\pi^2(\cos 2\pi x + \sin 2\pi x), \qquad (g')^2 = 4\pi^2(1 - \sin 4\pi x).$$

Then $u'' = e^g\big[(g')^2 + g''\big]$, so

$$-u'' = 4\pi^2 e^g\big[\sin 4\pi x - \cos 2\pi x - \sin 2\pi x - 1\big] = f, \tag{39}$$

and $u(0) = u(1) = e^0 - 1 = 0$. Numerically $\max|-u'' - f| = 3.4 \times 10^{-13}$ and $|u(0)| + |u(1)| = 2.2 \times 10^{-16}$.

### Convergence table

| $M$ | $h$ | $\|u - U\|_{L^2}$ | rate | $\|u - U\|_E$ | rate |
|---|---|---|---|---|---|
| 24 | 4.0000e-02 | 1.1386e-03 | — | 1.8042e-01 | — |
| 49 | 2.0000e-02 | 1.4056e-04 | 3.02 | 4.5288e-02 | 1.99 |
| 99 | 1.0000e-02 | 1.7513e-05 | 3.00 | 1.1334e-02 | 2.00 |
| 199 | 5.0000e-03 | 2.1874e-06 | 3.00 | 2.8341e-03 | 2.00 |
| 399 | 2.5000e-03 | 2.7336e-07 | 3.00 | 7.0857e-04 | 2.00 |
| 799 | 1.2500e-03 | 3.4169e-08 | 3.00 | 1.7715e-04 | 2.00 |

### Conclusion

$$\|u - U\|_E = O(h^{2.00}), \qquad \|u - U\|_{L^2} = O(h^{3.00}),$$

matching the CG(2) estimates of Lecture 10, $\|u - U\|_E \le c\|h^2 u'''\|$ and $\|u - U\| \le c\|h^3 u'''\|$. Each additional polynomial degree buys one power of $h$ in both norms, and the $L^2$ rate remains one order ahead of the energy rate.

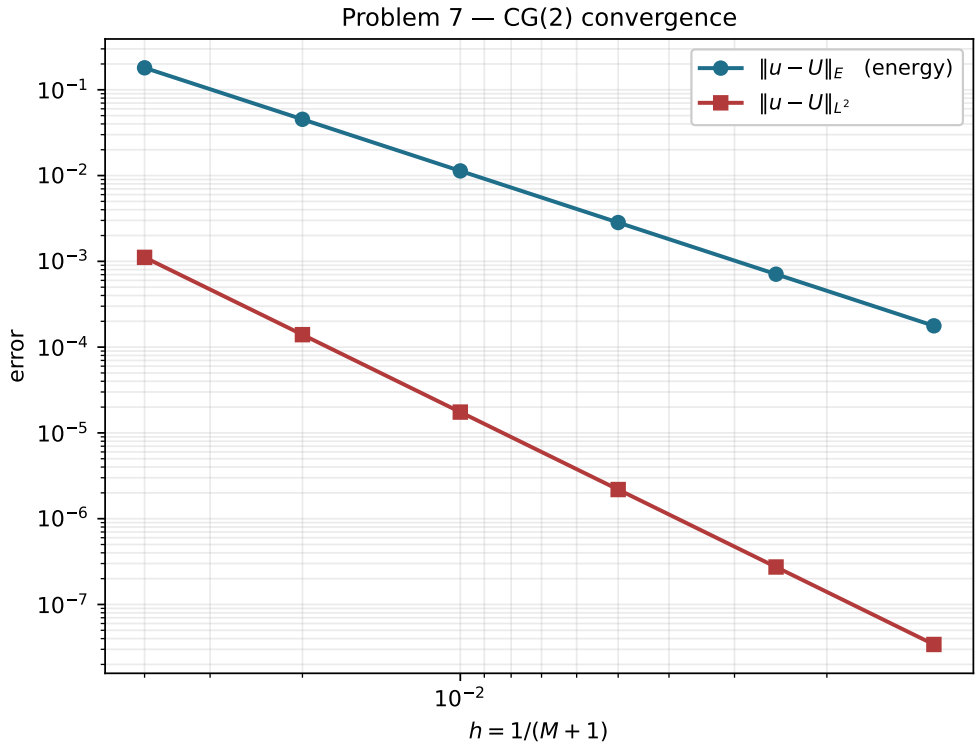



Figure 3: Problem 7. CG(2) convergence: energy-norm and $L^2$-norm errors against the mesh size $h = 1/(M+1)$, for $M = 24, 49, 99, 199, 399, 799$. Measured slopes 2.00 and 3.00.

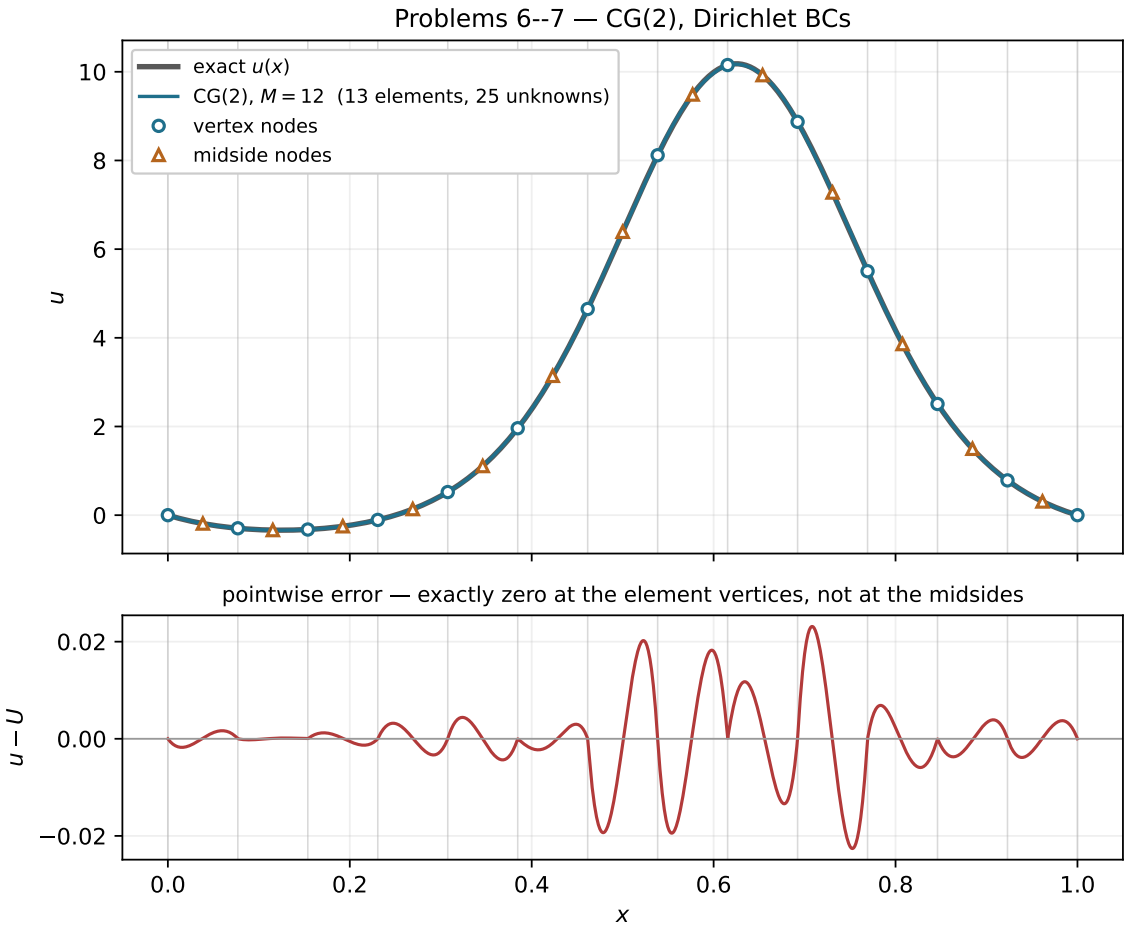



Figure 4: Problem 7. **Top:** exact solution and the CG(2) solution at $M = 12$; circles are vertex nodes and triangles midside nodes. **Bottom:** the pointwise error $u - U$.